## 1. Problem Statement

Navigating intellectual property and regulatory requirements for traditional medicine, particularly Ayurveda, Siddha and Unani (ASU), is challenging because multiple national laws and international treaties may apply simultaneously. Practitioners, researchers, micro, small and medium enterprises (MSMEs) and startups often struggle to identify the applicable legal provisions and understand how different regulatory frameworks affect their formulations. Questions involving patentability and traditional knowledge alongside biodiversity-related obligations and product classification can require careful examination of several authoritative documents. General-purpose AI systems may produce inaccurate legal explanations or unsupported citations which can mislead users during regulatory research. Furthermore, lengthy statutory documents and complex legal terminology make it difficult for non-specialists to locate and interpret relevant information efficiently.

**IP-SAKTI Sahayak** addresses these challenges through a jurisdiction-aware and formulation-specific Retrieval-Augmented Generation (RAG) system. It uses a curated regulatory corpus and semantic retrieval to identify relevant legal evidence before generating structured guidance. By validating source references and physical PDF page mappings, the system aims to make regulatory information more accessible and traceable. It also identifies insufficient evidence rather than presenting unsupported information as established legal guidance.

## 2. Aim of the Mini Project

The aim of IP-SAKTI Sahayak is to develop and evaluate a source-grounded Retrieval-Augmented Generation (RAG) system that:

- **Corpus Curation and Preprocessing:** Organizes authoritative regulatory documents into structured records while preserving original legal text and verifying source metadata and physical PDF page references.
- **Semantic Retrieval and Response Generation:** Uses MiniLM embeddings and FAISS similarity search to identify relevant legal passages and provide contextual evidence to the configured Gemini model for generating structured regulatory responses.
- **Evidence Validation and System Evaluation:** Applies deterministic checks to source identifiers and exact quotations alongside page mappings and response structure. The system is evaluated through automated tests and retrieval benchmarks to assess evidence traceability and its handling of incomplete or unsupported queries.

Together, these objectives establish a structured and auditable data science foundation for intellectual property and Ayurveda-related regulatory guidance. The mini-project focuses on the core RAG pipeline and its evidence-validation mechanisms before completing the integration of the existing frontend and backend workflows.

## 3. Corpus

The foundation of IP-SAKTI Sahayak is a curated, multi-source statutory corpus rather than a conventional single-table CSV dataset. Legal text cannot be scraped indiscriminately; every document must represent an official statutory enactment or multilateral treaty issued by a recognized governmental authority.

**Table 1: Authoritative Regulatory Source Documents in Pilot Corpus**

| Document Identifier | Document Title | Issuing Authority / Portal | Physical Pages |
|---|---|---|---|
| IND-BIO-ABS-001 | Biological Diversity (Access to Biological Resources and ABS) Regulations, 2025 | National Biodiversity Authority (NBA) | 7 |
| IND-BIO-ACT-001 | The Biological Diversity Act, 2002 | Legislative Department, India Code | 9 |
| IND-COPYRIGHT-001 | The Copyright Act, 1957 | Ministry of Education / India Code | 14 |
| IND-DESIGN-001 | The Designs Act, 2000 | CGPDTM, IP India / India Code | 7 |
| IND-GI-001 | Geographical Indications of Goods Act, 1999 | CGPDTM, IP India / India Code | 4 |
| IND-PAT-ACT-001 | The Patents Act, 1970 (as amended) | CGPDTM, IP India | 11 |
| IND-PPVFR-001 | Protection of Plant Varieties & Farmers' Rights Act, 2001 | Ministry of Agriculture / India Code | 15 |
| IND-REG-DCA-001 | The Drugs and Cosmetics Act, 1940 | Ministry of Health & Family Welfare / CDSCO | 12 |
| IND-TM-001 | The Trade Marks Act, 1999 | CGPDTM, IP India / India Code | 8 |
| INT-WIPO-GRATK-001 | WIPO Treaty on Intellectual Property, Genetic Resources and Associated TK (2024) | World Intellectual Property Organization | 5 |
| INT-WTO-TRIPS-001 | Agreement on Trade-Related Aspects of Intellectual Property Rights (TRIPS) | World Trade Organization | 8 |

## 4. Theory / Background

### 4.1 Regulatory Knowledge Retrieval

IP-SAKTI Sahayak uses Retrieval-Augmented Generation (RAG) to identify relevant legal provisions from authoritative documents and generate clear regulatory guidance supported by traceable citations.

### 4.2 Corpus Preprocessing and Chunking

The system validates source metadata and preserves original legal text. It divides legal records into chunks of up to 220 tokens with a 32-token overlap while retaining source identifiers and physical PDF page mappings.

### 4.3 Semantic Embeddings and FAISS Retrieval

The all-MiniLM-L6-v2 model converts legal text into normalized 384-dimensional embeddings. FAISS IndexFlatIP ranks relevant passages using cosine similarity to retrieve evidence for response generation.

### 4.4 Gemini Generation and Evidence Validation

The system uses Gemini to generate structured regulatory responses from retrieved evidence. Deterministic validators check source identifiers and exact text matches alongside physical page mappings and response structure. Scope guards identify incomplete answers and insufficient evidence.

## 5. Data Preprocessing

Data preprocessing was performed to validate the regulatory corpus and prepare legal text for semantic retrieval. The pilot contains 11 PDFs covering 100 physical pages and 357 structured legal records. Python-based validation preserves original statutory text and maintains source traceability throughout preprocessing.

### 5.1 Data Cleaning and Validation

- Validated mandatory metadata and physical PDF page mappings.
- Verified source integrity using SHA-256 hashes.
- Confirmed that all 357 records contain text and unique record identifiers.
- Preserved seven legitimate repeated legal phrases to avoid losing statutory context.

### 5.2 Chunking and Feature Transformation

Legal records were divided into 546 retrieval chunks using a maximum of 220 tokens and a 32-token overlap. Each chunk retains its source identifiers and physical page references. The all-MiniLM-L6-v2 model generates normalized 384-dimensional embeddings through ONNX Runtime for semantic retrieval.

### 5.3 Final Corpus Representation

The final retrieval representation contains the following fields.

**Table 2: Final Retrieval Chunk Fields**

| Field | Purpose |
|---|---|
| chunk_id | Unique chunk identification |
| record_id and document_id | Source provenance and document identification |
| text | Original legal text passage |
| tokens_without_specials | Chunk length for context management |
| pdf_pages | Physical PDF page references |
| Character offsets and SHA-256 hash | Text integrity and source traceability |
| required_context_unit_ids | Related provisions and required context |

**Table 3: Preprocessing Results**

| Metric | Final Result |
|---|---|
| Source PDFs | 11 |
| Physical pages | 100 |
| Validated records | 357 |
| Native source span proofs | 1,078 |
| Retrieval chunks | 546 |
| Embedding dimensions | 384 |
| Duplicate record IDs | 0 |

**Result**: The preprocessing pipeline produced a validated corpus of 357 legal records and 546 retrieval chunks with source traceability and normalized semantic embeddings ready for FAISS indexing.

## 6. Data Analysis and Visualization

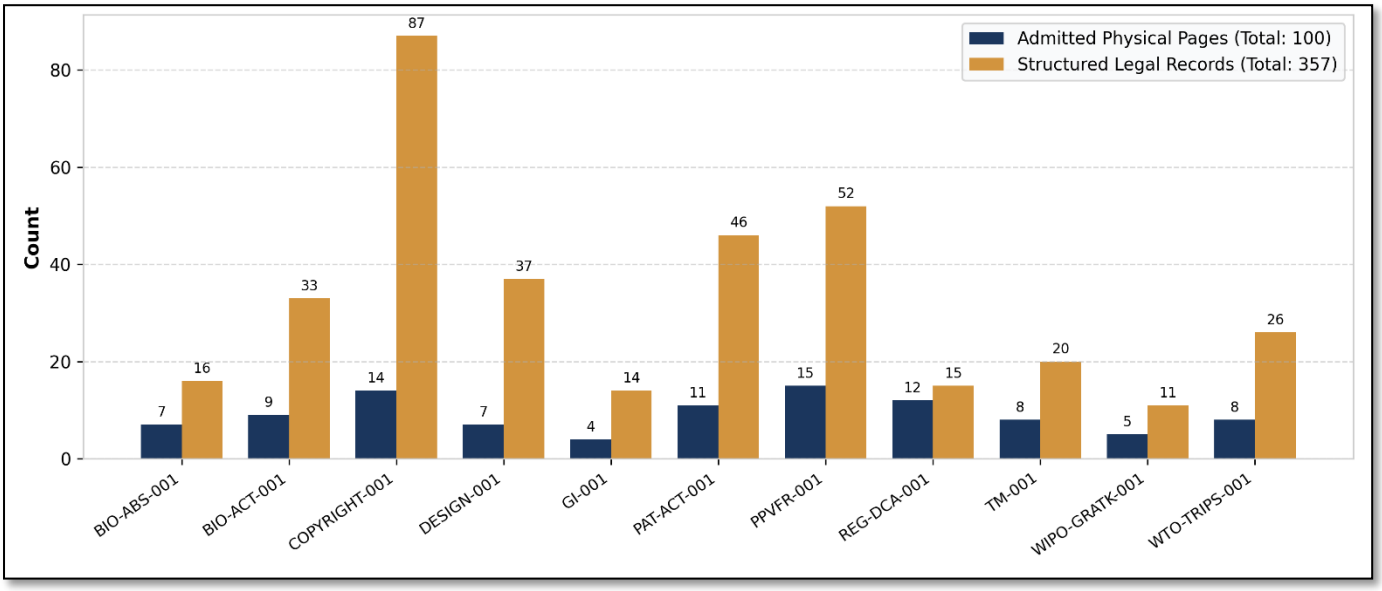



**Figure 6.1: Admitted Physical Pages and Structured Records Across Source Documents**

**Observation**: The pilot corpus consists of 11 legal documents covering 100 physical PDF pages and 357 structured legal records. The chart shows how the records and pages are distributed across the source documents.

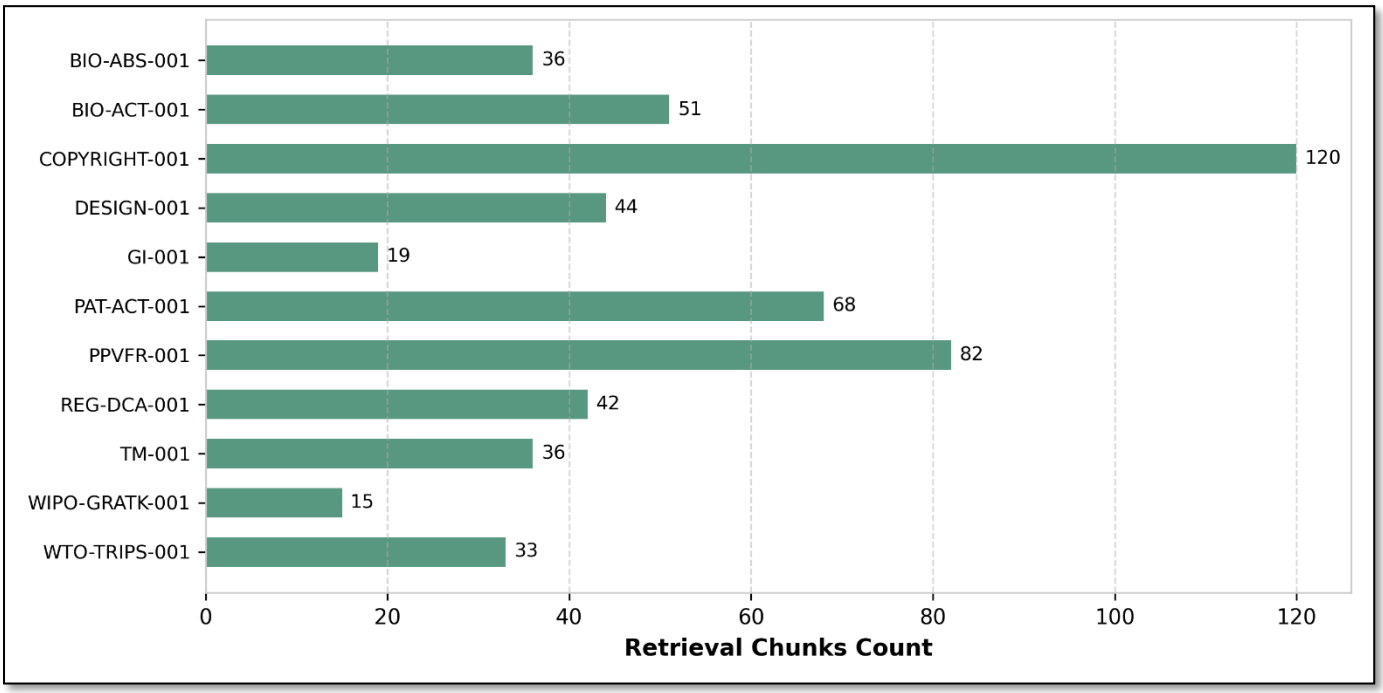



**Figure 6.2: Number of Retrieval Chunks per Source Document**

**Observation**: The preprocessing pipeline generated 546 retrieval chunks from the admitted legal records. The distribution varies across documents, with the Copyright Act contributing the highest number of chunks at 120. These chunks provide smaller text units for semantic retrieval.

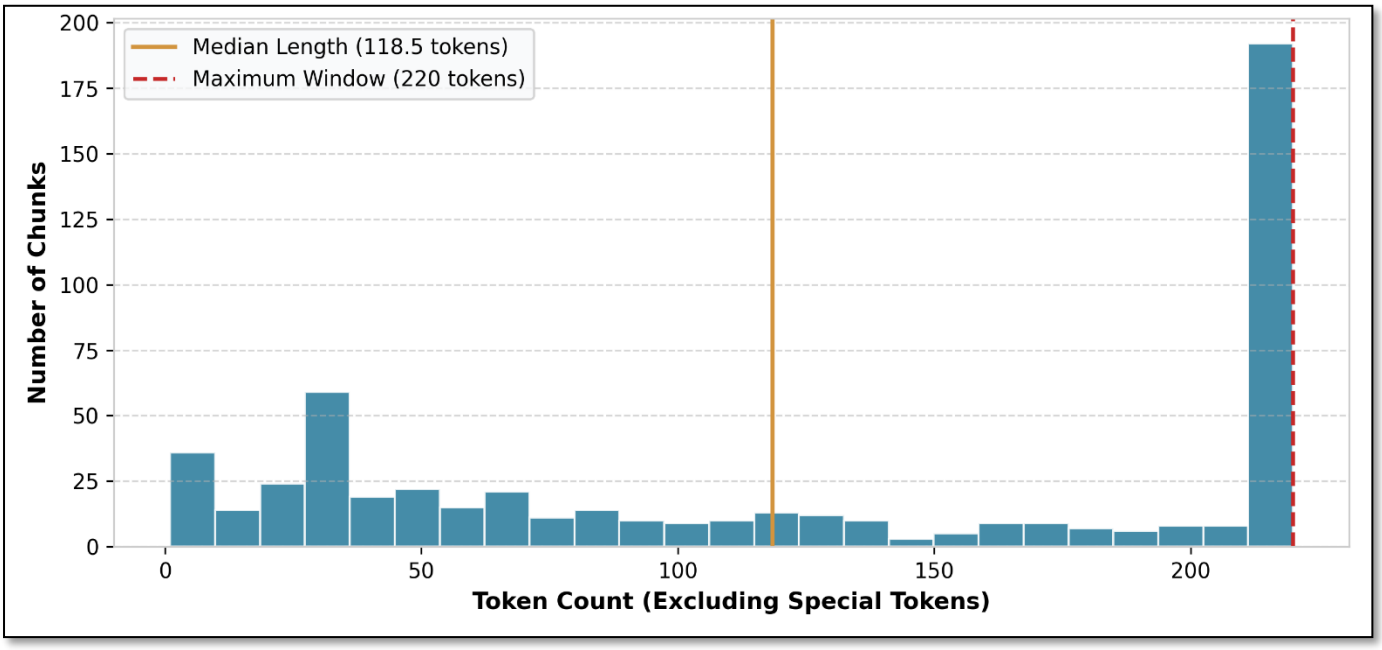



**Figure 6.3: Chunk Token-Length Distribution**

**Observation**: The chart shows the token-length distribution of 546 retrieval chunks. Chunk lengths range up to the configured maximum of 220 tokens, with a median length of 118.5 tokens. This helps demonstrate how the legal text is organized into manageable units for embedding and retrieval.

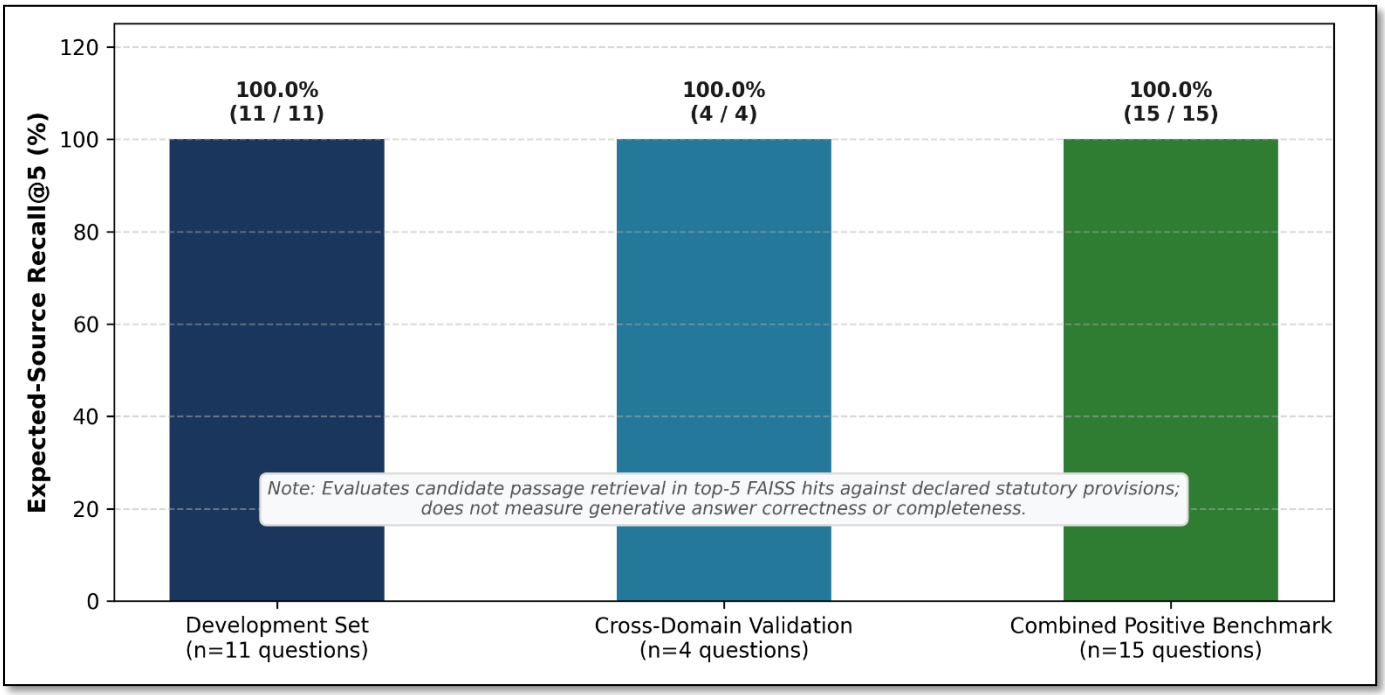



**Figure 6.4: Retrieval Effectiveness on the Regulatory Evaluation Set**

**Observation**: The retrieval evaluation achieved 100% expected-source retrieval across 15 tested questions, comprising 11 development questions and four cross-domain validation questions. The result indicates that the expected legal sources appeared among the top five retrieved results for the evaluated queries. It does not measure the accuracy or completeness of the generated legal answer.

## 7. Data Science Technique / Model

### 7.1 Retrieval-Augmented Generation (RAG)

IP-SAKTI Sahayak combines semantic retrieval with generative AI to provide source-cited regulatory guidance. The pipeline uses MiniLM embeddings and FAISS to retrieve relevant legal provisions before passing the evidence to Gemini for structured response generation.

### 7.2 Embedding and Semantic Retrieval

The all-MiniLM-L6-v2 model generates normalized 384-dimensional embeddings through ONNX Runtime. FAISS IndexFlatIP performs similarity-based retrieval across 546 vectors and expands relevant context to include related statutory provisions.

### 7.3 Gemini Generation and Evidence Validation

The system uses Google's Gemini API (gemini-3.8-flash) to generate structured responses containing summaries and legal references. Deterministic validators check exact quotations and source identifiers alongside physical PDF page mappings and response structure. Unsupported queries are handled through insufficient-evidence responses.

### 7.4 Model and System Evaluation

The system was evaluated using automated tests and retrieval benchmarks alongside historical model outputs and a live Gemini request.

**Table 4: Model and System Evaluation Results**

| Evaluation Metric | Result |
|---|---|
| Offline tests | 166/166 passed |
| Expected-source Recall@5 | 15/15 queries (100%) |
| Historical citation checks | 20/20 passed |
| Current Gemini generation | One live request with citation validation passed |

**Result:** effective source retrieval and citation validation on the evaluated cases. Answer completeness requires further evaluation and full website integration remains pending.

## 8. Front-End / User Interface

IP-SAKTI Sahayak provides a web-based interface for accessing intellectual property guidance and Ayurveda-related regulatory information. The interface combines user authentication and guided workflows with a technology architecture designed for source-based legal information retrieval.

### 8.1 Home Page and Navigation

The homepage introduces IP-SAKTI Sahayak and its purpose of providing verified intellectual property guidance. The navigation bar provides access to Guidance, Formulation Classifier and Resource Library alongside Login and Register options.

### 8.2 User Authentication

The login and registration interface allows users to access their workspace. Authentication supports a personalized application experience for users accessing the available features.

### 8.3 Regulatory Guidance

Users select India or International as the jurisdiction and choose a regulatory domain such as Patents, Biodiversity or ASU Regulatory. The Quick Inquiries dropdown provides predefined questions while the input field allows users to submit queries in natural language. Reset and submission controls support successive inquiries in a chat-style interface.

### 8.4 Formulation Classifier

The classifier uses a four-step questionnaire covering textual origin and intended use alongside processing method and excipients or delivery systems. A progress indicator and Back and Next Step controls guide users through the questions before they submit their selections for formulation assessment.

### 8.5 System Architecture and Technology Stack:

The workflow connects user queries with trusted legal sources through Python and FAISS-based semantic retrieval. The configured Gemini API generates structured responses while validation mechanisms check citations and source references. The technology stack includes React and JavaScript alongside Tailwind CSS and Node.js with Express and MongoDB. Python and FAISS support the retrieval pipeline while Gemini API provides the current generation capability.

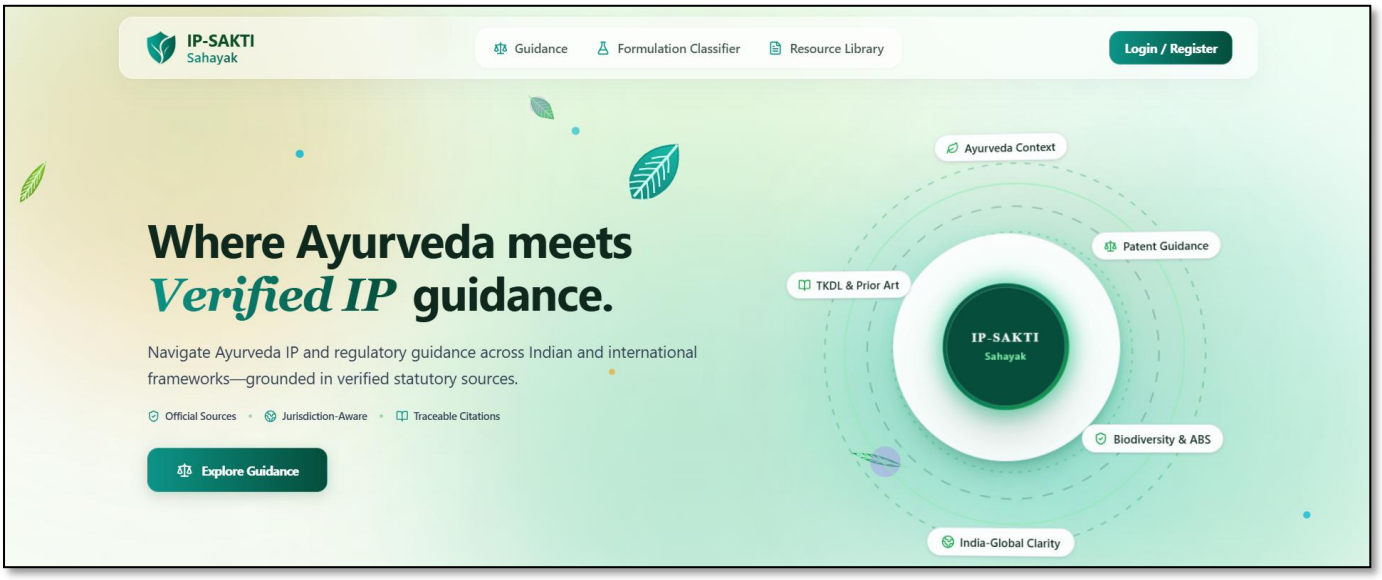



**Figure 8.1: IP-SAKTI Sahayak Home Page**

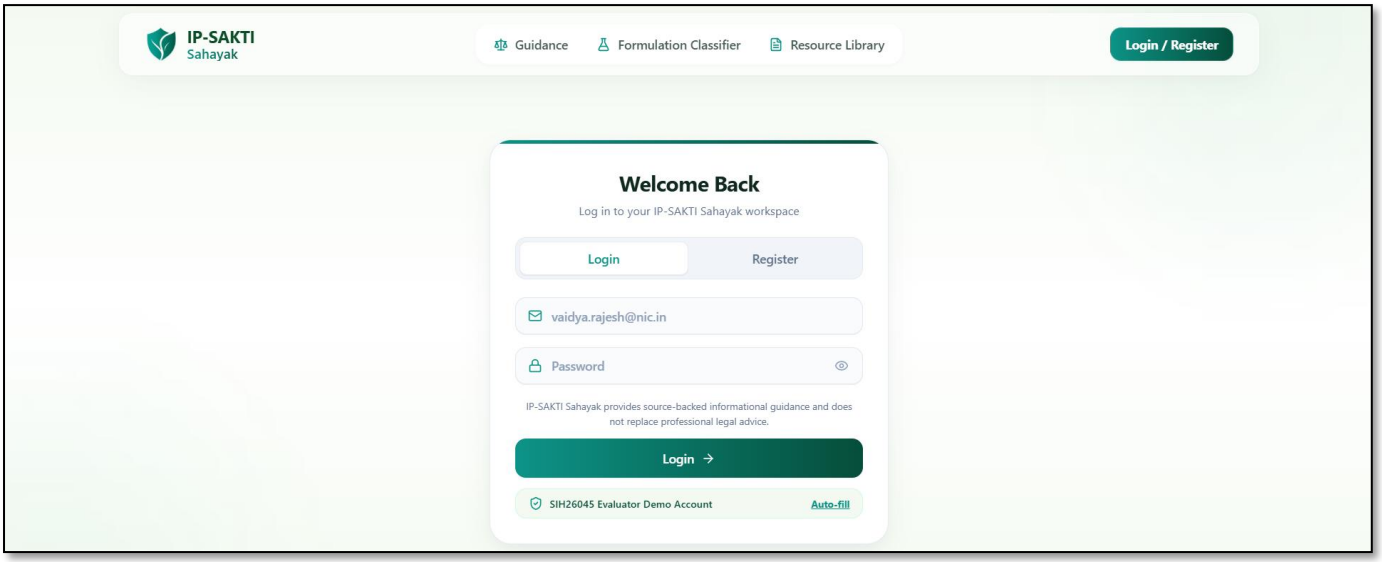



**Figure 8.2: User Sign-Up and Login Interface**




**Figure 8.3: Regulatory Guidance Interface**

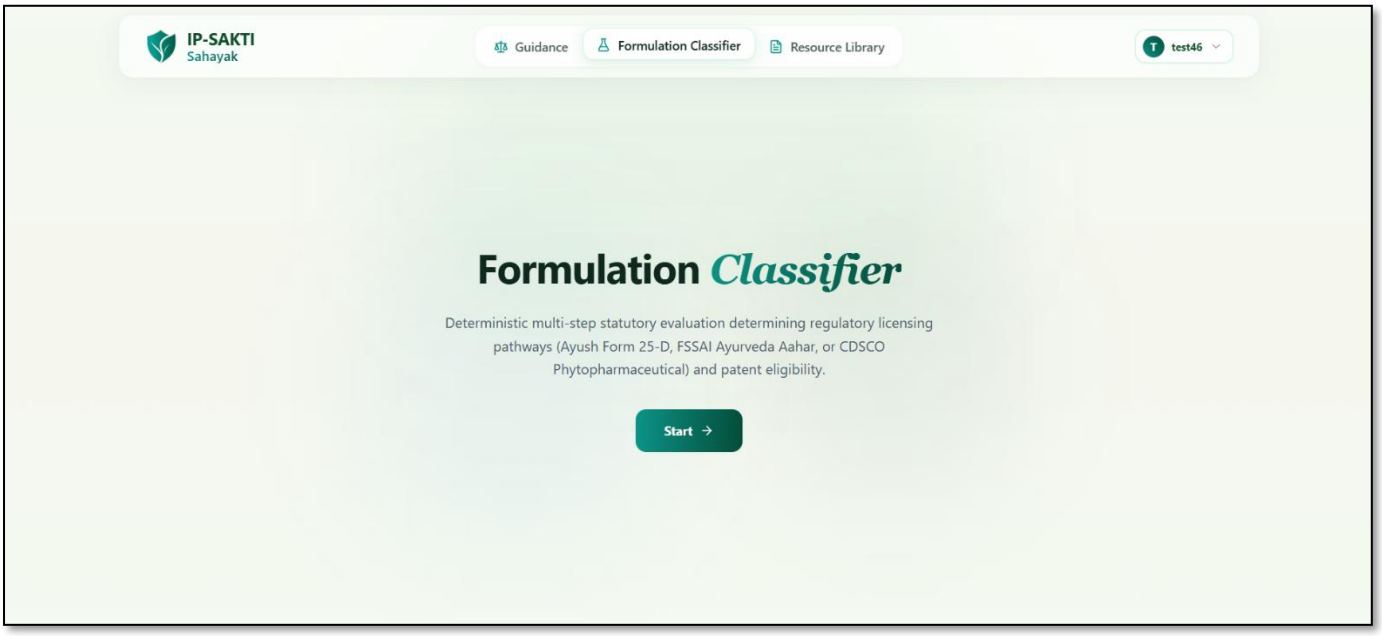



**Figure 8.4: Formulation Classifier Interface**

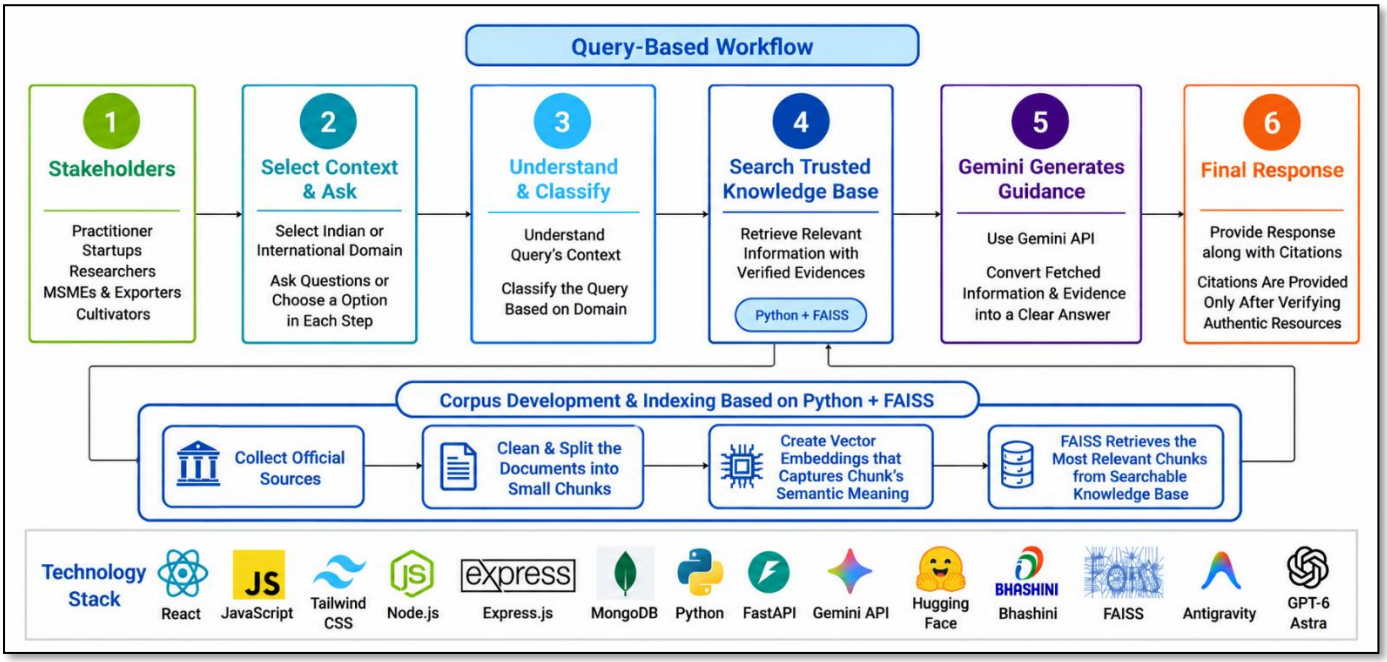



**Figure 8.5: Query-Based System Architecture and Technology Stack**

## 9. Complete Code

This section presents selected code excerpts from the core implementation of IP-SAKTI Sahayak. The code demonstrates corpus validation and preprocessing alongside semantic embedding generation and retrieval-based response validation.

### 9.1 Source Validation (model_1.O/src/ipsakti/corpus.py)

```
def record_pages(record):
    pages = []

    for key in ("pdf_page", "pdf_pages"):
        value = record.get(key)

        if isinstance(value, list):
            pages.extend(value)
        elif isinstance(value, int):
            pages.append(value)

    return sorted(set(
        page for page in pages
        if type(page) is int
    ))


def indexing_reasons(record, workflow):
    reasons = []

    if workflow.get("indexing_approved") is not True:
        reasons.append("canonical_indexing_gate_closed")

    if not record.get("document_id"):
        reasons.append("missing_document_id")

    if not record.get("record_id"):
        reasons.append("missing_id")

    if not record.get("text", "").strip():
        reasons.append("missing_text")

    if not record_pages(record):
        reasons.append("missing_physical_pages")

    return reasons
```

**Explanation:** This code validates essential source metadata and physical PDF page references before allowing a legal record to proceed through the indexing workflow.

**9.2 Embedding Generation (model_1.O/src/ipsakti/models.py)**

```
def encode(self, texts):
    if not texts:
        raise ValueError("Cannot embed an empty dataset")

    vectors = []

    for offset in range(0, len(texts), 32):
        batch = self.tokenizer.encode_batch(
            texts[offset:offset + 32]
        )

        inputs = {
            "input_ids": np.array(
                [item.ids for item in batch],
                dtype=np.int64
            ),
            "attention_mask": np.array(
                [item.attention_mask for item in batch],
                dtype=np.int64
            ),
            "token_type_ids": np.array(
                [item.type_ids for item in batch],
                dtype=np.int64
            )
        }

        feeds = {
            item.name: inputs[item.name]
            for item in self.session.get_inputs()
        }

        hidden = self.session.run(None, feeds)[0]
        mask = inputs["attention_mask"][..., None]

        pooled = (hidden * mask).sum(axis=1)
        pooled /= mask.sum(axis=1).clip(min=1)

        pooled = pooled.astype("float32")
        pooled /= np.linalg.norm(
            pooled, axis=1, keepdims=True
        ).clip(min=1e-12)

        vectors.append(pooled)

    return np.ascontiguousarray(
        np.concatenate(vectors)
    )
```

**Explanation:** The embedding pipeline uses attention-masked mean pooling and L2 normalization to produce 384-dimensional vectors suitable for semantic similarity search.

**9.3 FAISS Semantic Retrieval (model_1.O/src/ipsakti/retrieval.py)**

```
def search(self, question, k=5, min_similarity=0.25,
           max_characters=12000):

    scores, indices = self.index.search(
        self.embedder.encode([question]),
        min(k, len(self.chunks))
    )

    results = []
    seen = set()

    for score, index in zip(scores[0], indices[0]):
        if index >= 0 and score >= min_similarity:
            chunk = {
                **self.chunks[int(index)],
                "similarity": float(score),
                "retrieval_role": "direct_hit"
            }

            results.append(chunk)
            seen.add(chunk["chunk_id"])

    return results
```

**Explanation:** FAISS retrieves the most similar legal passages and applies a minimum similarity threshold. The complete implementation also expands required context units when assembling evidence for the answer.

**9.4 Citation and Evidence Validation (model_1.O/src/ipsakti/retrieval.py)**

```
def validate_answer(generation, evidence):
    report = generation.get("parsed")

    mapping = {
        chunk["chunk_id"]: chunk
        for chunk in evidence
    }

    if not isinstance(report, dict):
        raise Blocked("Invalid answer schema")

    for claim in report.get("claims", []):
        for citation in claim.get("citations", []):
            chunk_id = citation.get("chunk_id")

            if chunk_id not in mapping:
                raise Blocked("Invented citation ID detected")

            chunk = mapping[chunk_id]
            quote = citation.get("quote", chunk["text"])

            if quote not in chunk["text"]:
```

```
                raise Blocked("Quotation is not an exact retrieved span")

            citation["pdf_pages"] = chunk["pdf_pages"]
            citation["document_id"] = chunk["document_id"]

    return {
        **report,
        "citation_checks_passed": True
    }
```

**Explanation:** The validation process checks whether cited chunk identifiers exist and quoted text matches the retrieved evidence. Verified document identifiers and physical page references are attached to citations.

### 9.5 RAG Service Bridge (model_1.O/scripts/serve_rag.py)

```
def dispatch(workflow, payload, cache, timeout=180):
    gate = preflight(workflow, payload)

    if gate["status"] != "ready":
        return envelope(
            workflow,
            payload,
            gate["status"],
            gate["message"]
        )

    try:
        from ipsakti.service import (
            handle_request,
            public_response
        )

        response = handle_request(
            root,
            workflow,
            payload,
            cache=pathlib.Path(cache) if cache else None
        )

        return public_response(
            response,
            root / ".runtime/audit"
        )

    except Exception as exc:
        return envelope(
            workflow,
            payload,
            "service_unavailable",
            f"Native request failed: {exc}"
        )
```

**Explanation:** The service bridge performs request preflight checks and forwards valid requests to the RAG engine. It returns a structured response or a controlled service-unavailable status when processing fails.

## 10. Output Screenshots

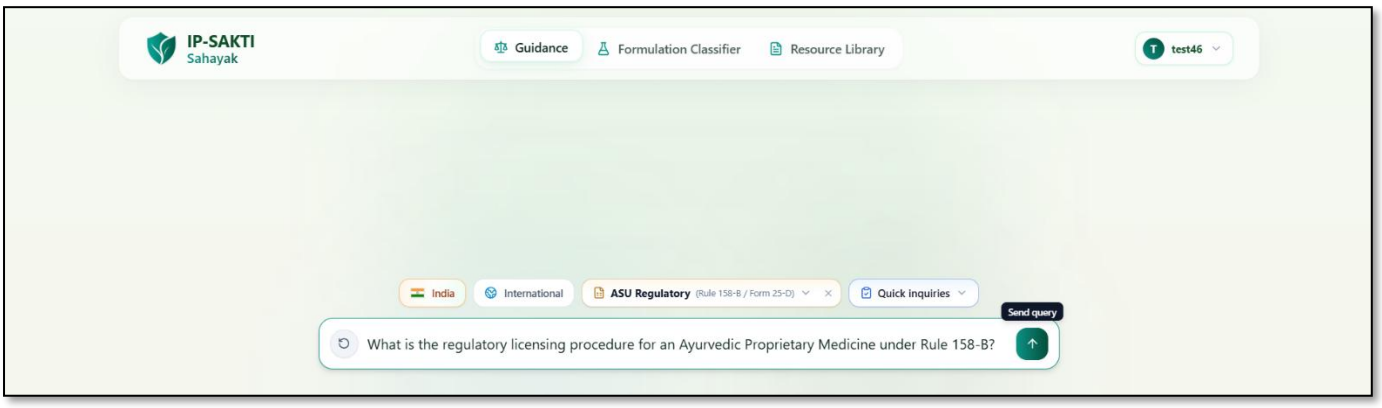



**Figure 10.1 — Regulatory Guidance Query Interface**

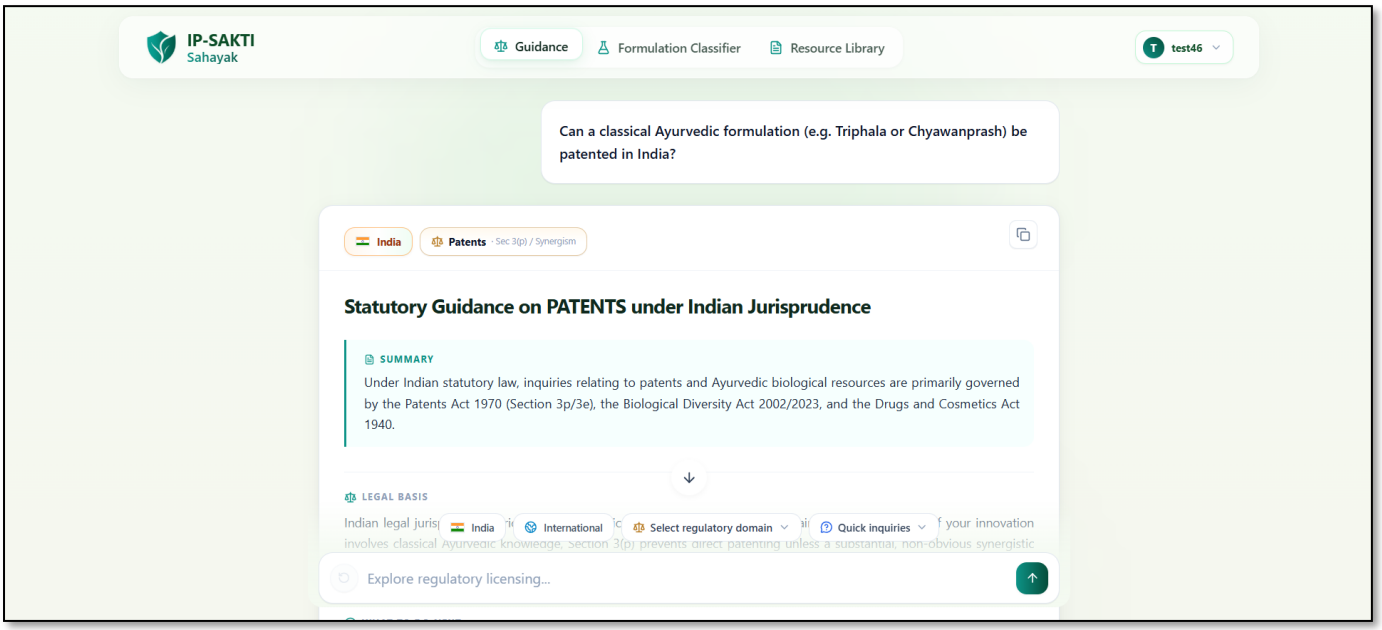



**Figure 10.2 — Structured Guidance Response**

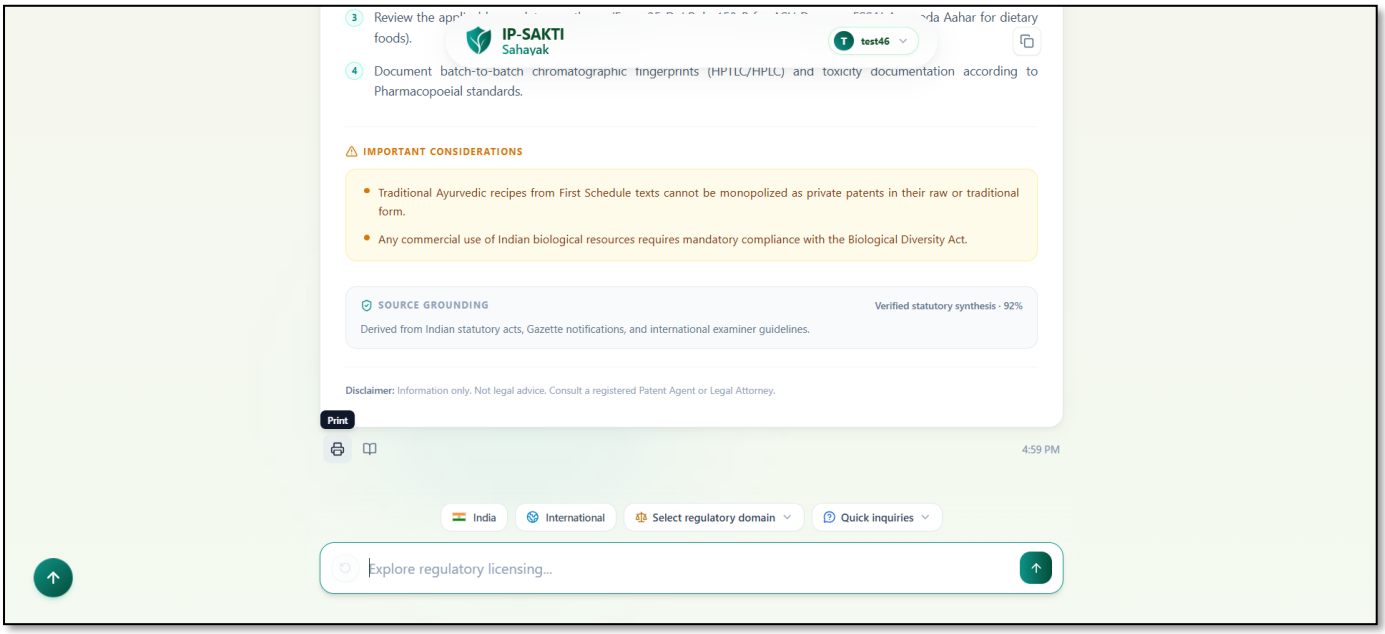
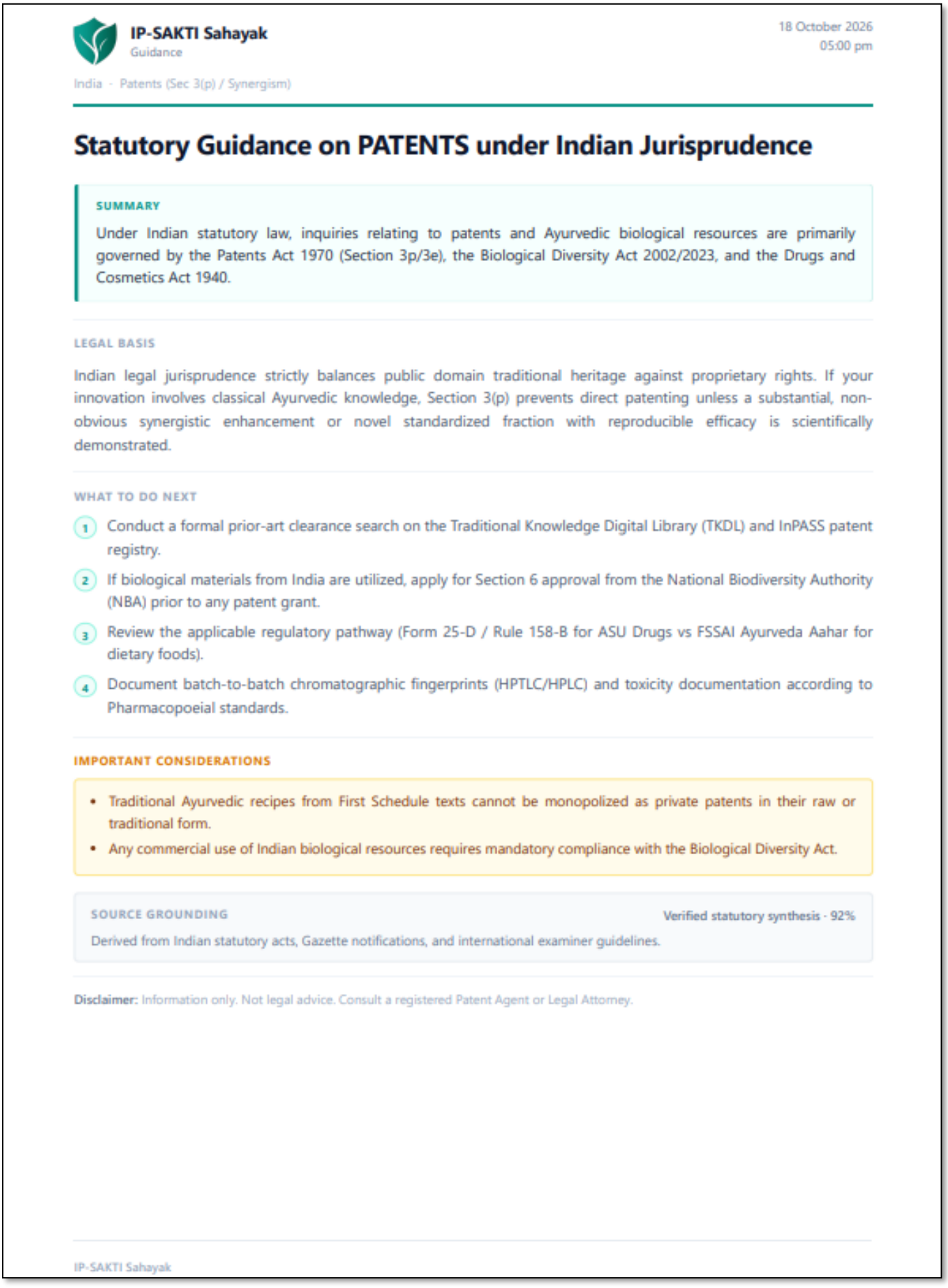



**Figure 10.3 — Guidance Details and Source Grounding**



IP-SAKTI Sahayak
Guidance

18 October 2026
05:00 pm

India · Patents (Sec 3(p) / Synergism)

# Statutory Guidance on PATENTS under Indian Jurisprudence

SUMMARY

Under Indian statutory law, inquiries relating to patents and Ayurvedic biological resources are primarily governed by the Patents Act 1970 (Section 3p/3e), the Biological Diversity Act 2002/2023, and the Drugs and Cosmetics Act 1940.

LEGAL BASIS

Indian legal jurisprudence strictly balances public domain traditional heritage against proprietary rights. If your innovation involves classical Ayurvedic knowledge, Section 3(p) prevents direct patenting unless a substantial, non-obvious synergistic enhancement or novel standardized fraction with reproducible efficacy is scientifically demonstrated.

WHAT TO DO NEXT

1. Conduct a formal prior-art clearance search on the Traditional Knowledge Digital Library (TKDL) and InPASS patent registry.
2. If biological materials from India are utilized, apply for Section 6 approval from the National Biodiversity Authority (NBA) prior to any patent grant.
3. Review the applicable regulatory pathway (Form 25-D / Rule 158-B for ASU Drugs vs FSSAI Ayurveda Aahar for dietary foods).
4. Document batch-to-batch chromatographic fingerprints (HPTLC/HPLC) and toxicity documentation according to Pharmacopoeial standards.

IMPORTANT CONSIDERATIONS

- Traditional Ayurvedic recipes from First Schedule texts cannot be monopolized as private patents in their raw or traditional form.
- Any commercial use of Indian biological resources requires mandatory compliance with the Biological Diversity Act.

SOURCE GROUNDING — Verified statutory synthesis - 92%

Derived from Indian statutory acts, Gazette notifications, and international examiner guidelines.

**Disclaimer:** Information only. Not legal advice. Consult a registered Patent Agent or Legal Attorney.

IP-SAKTI Sahayak

**Figure 10.4 — Printable Guidance Report**

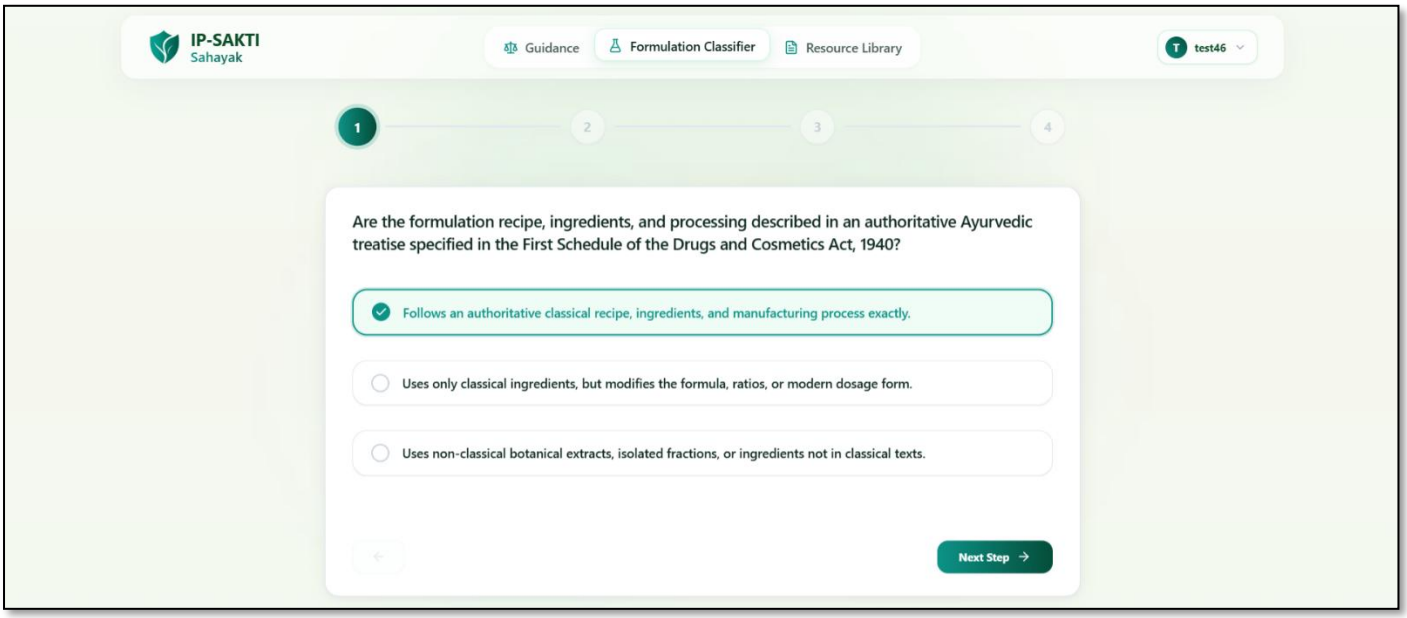



**Figure 10.5: Textual Origin Selection**

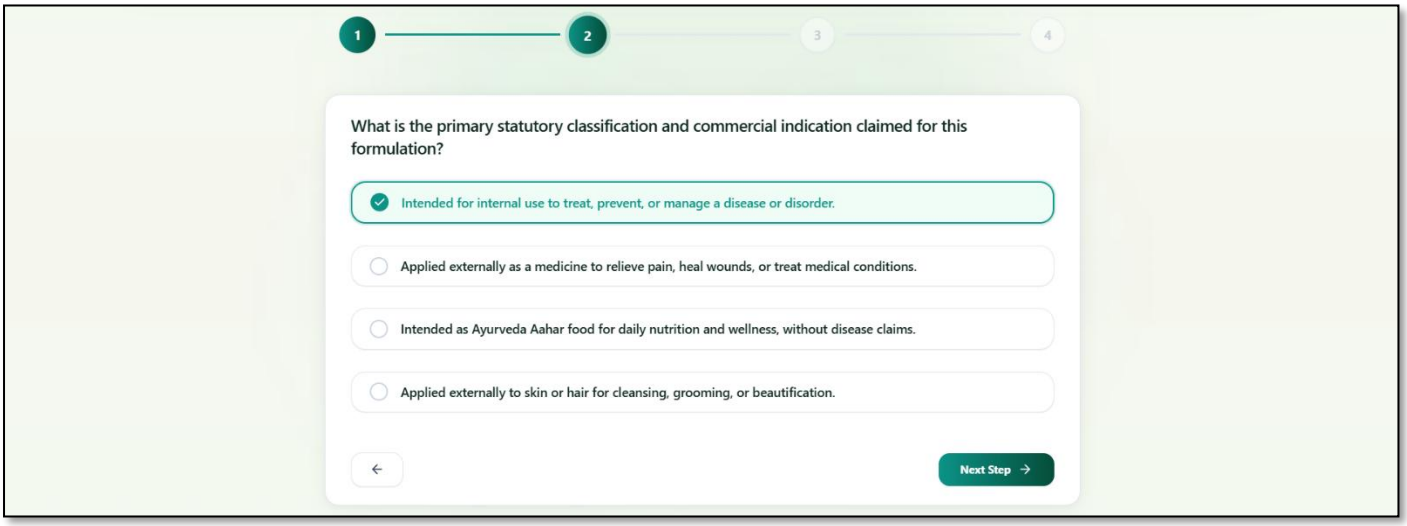



**Figure 10.6: Intended Use Selection**

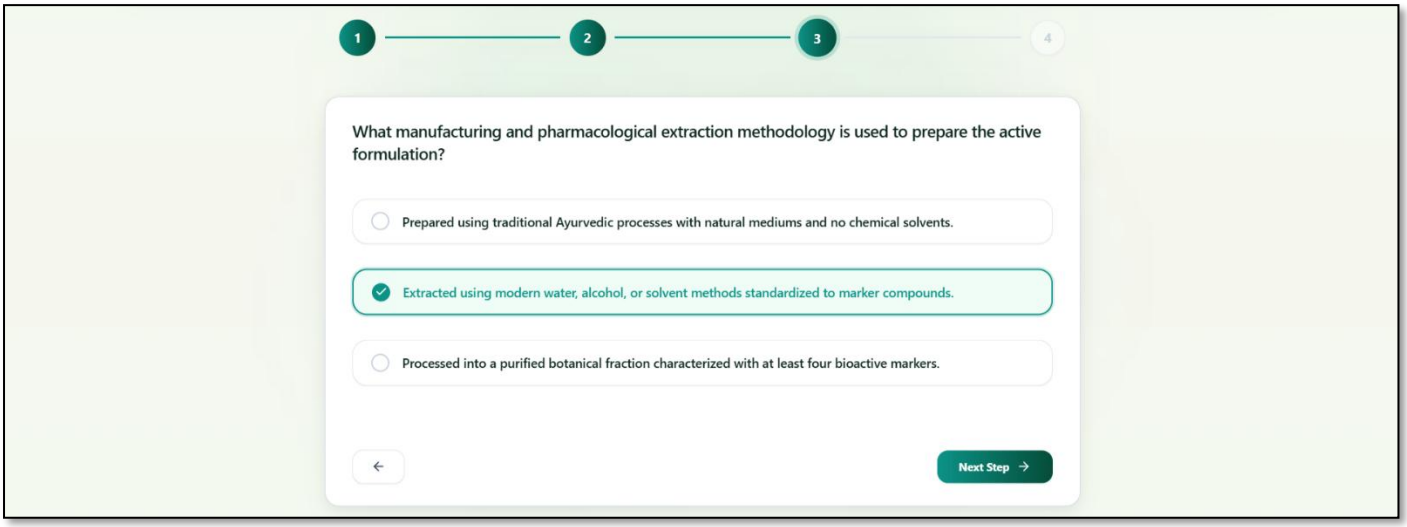



**Figure 10.7: Processing and Extraction Method**

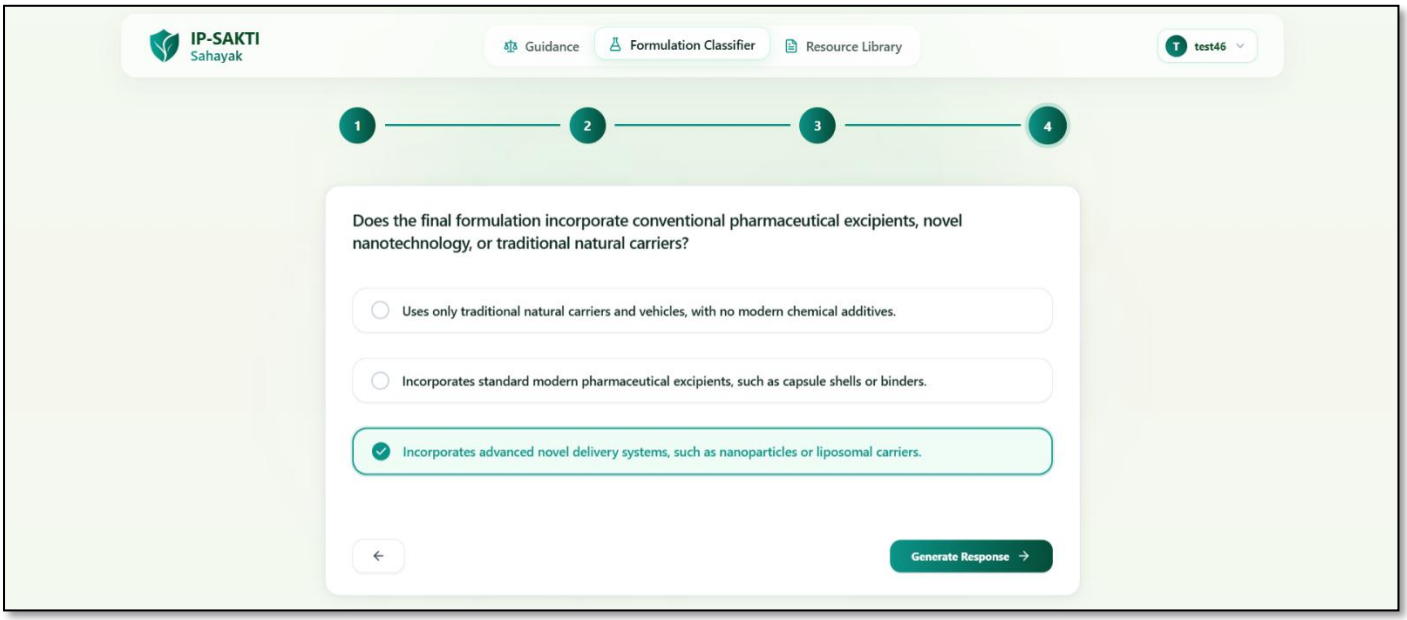



**Figure 10.8: Excipients and Delivery System**

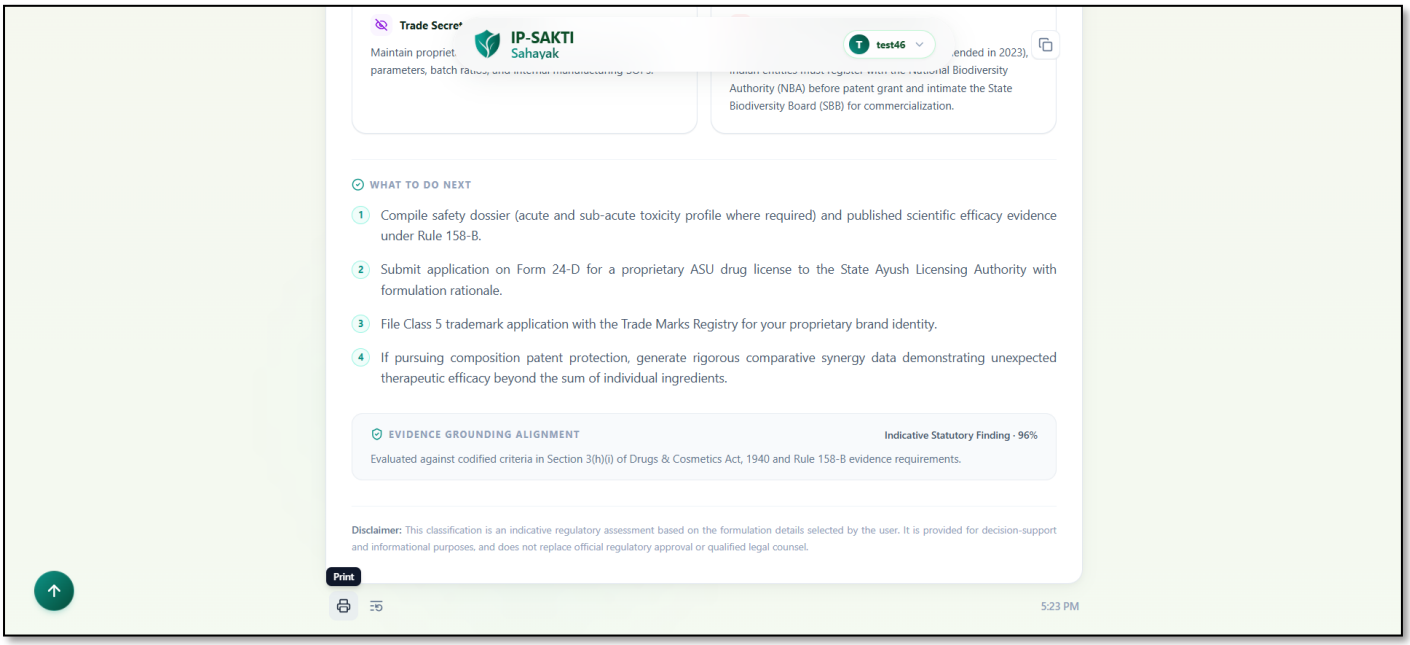



**Figure 10.9: Classification Results Dossier**




**Figure 10.10: Printable Assessment Report**

## 11. Observations

1. The interface allows users to select India or International and choose a regulatory domain before entering their question using the text input field and submission button.

2. The guidance response presents information in structured sections such as Summary and Legal Basis to help users understand the main findings and their supporting legal reasoning.

3. The response includes a What To Do Next section with numbered suggestions and an Important Considerations section highlighting relevant regulatory points.

4. Source-grounding information and a legal disclaimer are displayed to communicate the stated basis and limitations of the guidance. The print control allows users to prepare a printable report for reference.

5. The chat-style interface supports successive regulatory inquiries within the same conversation. The displayed response is demonstration content and requires verification after integration with the live RAG service.

6. The first step asks whether the formulation follows an authoritative Ayurvedic treatise specified in the First Schedule of the Drugs and Cosmetics Act, 1940. Users can select an exact classical formulation or indicate modifications or non-classical ingredients before proceeding.

7. The second step collects information about the formulation's intended use. Users can select internal therapeutic use and external application alongside Ayurveda Aahara for dietary purposes or topical cosmetic use.

8. The third step asks how the formulation is manufactured or extracted. Users can select traditional Ayurvedic preparation methods or indicate modern extraction techniques or purified botanical fractions.

9. The fourth step examines the formulation's carriers and delivery systems. Users can select traditional natural carriers or modern pharmaceutical excipients alongside advanced technologies such as nanoparticles and liposomal carriers. The Generate Response button submits the completed questionnaire for assessment.

10. The resulting dossier presents the formulation category and supporting explanation alongside regulatory pathways and intellectual property considerations. It also provides suggested next steps and evidence-grounding information to help users review the assessment.

11. The printable report consolidates the classification summary and legal basis with regulatory requirements and intellectual property considerations. It also includes recommended next steps and a disclaimer for convenient review and reference.

## 12. Conclusion

The IP-SAKTI Sahayak mini-project successfully develops a source-grounded Retrieval-Augmented Generation (RAG) pipeline for intellectual property and regulatory guidance in Ayurveda. Using a curated corpus of 11 legal documents and 100 physical PDF pages, the system combines data preprocessing and MiniLM embeddings with FAISS retrieval and Gemini-based response generation to provide structured guidance supported by traceable legal evidence.

The system achieved 166 passing offline tests and demonstrated successful source retrieval and citation validation on the evaluated cases. Although the core RAG pipeline has been developed and tested, complete frontend-backend integration and broader evaluation remain pending. Overall, the project demonstrates how data science and generative AI can support accessible and evidence-based regulatory guidance while acknowledging the limitations of the available corpus.

## 13. References

### 13.1 Statutory Legal Sources


- The Patents Act, 1970: IP India. https://ipindia.gov.in/resource/patents-resources-act
- The Biological Diversity Act, 2002: India Code. https://www.indiacode.nic.in/
- Biological Diversity (Access to Biological Resources and ABS) Regulations, 2025: National Biodiversity Authority. https://nbaindia.nic.in/
- The Drugs and Cosmetics Act, 1940: CDSCO. https://cdsco.gov.in/
- The Trade Marks Act, 1999: India Code. https://www.indiacode.nic.in/
- The Geographical Indications of Goods Act, 1999: India Code. https://www.indiacode.nic.in/
- The Protection of Plant Varieties and Farmers' Rights Act, 2001: India Code. https://www.indiacode.nic.in/
- The Copyright Act, 1957: India Code. https://www.indiacode.nic.in/
- The Designs Act, 2000: IP India. https://ipindia.gov.in/
- WIPO Treaty on Intellectual Property, Genetic Resources and Associated Traditional Knowledge, 2024: WIPO. https://www.wipo.int/meetings/en/doc_details.jsp?doc_id=631433
- Agreement on Trade-Related Aspects of Intellectual Property Rights (TRIPS): WTO. https://www.wto.org/english/docs_e/legal_e/31bis_trips_01_e.htm


### 13.2 Official Institutional and Research Portals


- Ministry of AYUSH: Government of India. https://ayush.gov.in/
- Pharmacopoeia Commission for Indian Medicine and Homoeopathy: https://pcimh.gov.in/
- Food Safety and Standards Authority of India: https://www.fssai.gov.in/
- Traditional Knowledge Digital Library: CSIR and Ministry of AYUSH. https://www.tkdl.res.in/
- Convention on Biological Diversity and Nagoya Protocol: https://www.cbd.int/

**13.3 Technical Documentation**


- FAISS: Efficient similarity search and clustering of dense vectors. https://github.com/facebookresearch/faiss
- Sentence-BERT: Sentence embeddings using Siamese BERT-Networks. https://www.sbert.net/
- ONNX Runtime: Machine-learning inference framework. https://onnxruntime.ai/
- Google Gemini API: Google AI for Developers. https://ai.google.dev/
- PyMuPDF: PDF text extraction and processing. https://pymupdf.readthedocs.io/
- NumPy: Numerical computing in Python. https://numpy.org/
- Matplotlib: Data visualization and plotting. https://matplotlib.org/
- Python: Official programming language documentation. https://docs.python.org/3/